

# Shortest XOR Straight-Line Programs for $3 \times 3$ Matrix Multiplication: Benchmark Description

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logic repository, <https://github.com/gsidebottom/logic>

**Abstract**—We contribute SAT instances deciding “do  $k$  XOR additions suffice?” for the output sides of rank-23 bilinear  $3 \times 3$  matrix-multiplication schemes. Each boundary instance is a live mathematical question: an UNSAT answer at the right  $k$  lower-bounds the integer additive complexity of a scheme class (an optimality certificate). The exact minima are now independently known — via the transposition principle, which just produced the first sub-56 scheme (55 additions) — so these instances carry known ground-truth answers yet remain hard for every solver we tried. The family’s AND-guarded parity structure defeats current XOR/Gaussian reasoning while remaining genuinely hard for plain CDCL: no tested configuration (kissat, CaDiCaL, CryptoMiniSat plain and native-XOR, Z3 word-level, portfolios with symmetry breaking) decides the calibrated boundary cells within 10–30 minutes.

## I. Problem origin

A bilinear scheme multiplying two  $3 \times 3$  matrices with 23 products [8] evaluates 23 products of linear forms and combines them into the 9 output entries. Its additive complexity — binary  $\pm$  additions in a straight-line program, negation free, no change of basis — has fallen rapidly: 60 [4], 59 [3], 58 [2], and the current record 56 [1]. Sun’s  $56 = 13 + 13 + 30$  splits into two provably optimal input sides and an output side of 30 additions computing the 9 output forms from the 23 products.

55 additions are now achieved (13+14+28 on class i19w225c4efh), by minimizing the output side exactly via the transposition principle — the record was 56 (Sun 2026). That also yields, for every scheme, the exact output-side minimum in milliseconds, so these instances now carry known ground-truth answers (the SAT/UNSAT boundary is independently pinned) while remaining hard for every SAT/SMT solver tried; the open target is now 54.

Any  $\mathbb{Z}$ -coefficient output-side program reduces mod 2 to an XOR straight-line program, so the decision problem

SLP( $k$ ): do  $k$  XOR additions suffice to compute the 9 given output forms (vectors in  $\text{GF}(2)^{23}$ ) from the 23 inputs?

yields sound lower bounds: UNSAT at  $k$  proves the integer output side needs  $\geq k+1$  additions, and UNSAT at  $k = 27$  on the right cells (§III) closes entire equivalence classes for 55. These instances continue the tradition of matrix-multiplication benchmarks in past competitions [5], with a twist of independent solver interest: the parity constraints

are AND-guarded, which prevents CryptoMiniSat’s Gaussian elimination from engaging, while plain CDCL faces genuine parity hardness.

## II. Encoding

SLP synthesis in the style of Fuhs–Schneider-Kamp [6], simplified by unit-vector inputs. For steps  $t = 1, \dots, k$ :

- Source selection. Step  $t$  selects exactly two sources among the 23 inputs and steps  $1..t-1$  (sequential counter for the exactly-two constraint).
- Values. Value bits are parity-defined:

$$x_{t,i} = \text{sel}_{t,\text{base}_i} \oplus \bigoplus_{j < t} (\text{sel}_{t,\text{step}_j} \wedge x_{j,i}).$$

Unit-vector inputs make the base contribution a single literal. The AND-guarded parities are materialized as Tseitin chains (plain CNF) or native x-lines (CryptoMiniSat extension); the generator emits both.

- Outputs. Each of the 9 forms must equal some step value (selector-guarded bit equalities; weight-1 forms may match an input).
- Symmetry breaking (optional, default on): all step values nonzero; adjacent independent steps (step  $t+1$  does not consume step  $t$ ) must have strictly lexicographically increasing values. Sound: swapping adjacent independent steps preserves validity, so every program normalizes by bubble sort, and padding above the minimum survives as a dependent chain. Dead-step elimination is deliberately not used: with it, satisfiability is not monotone in  $k$  (odd-length padding can force a dead step), which corrupts minimum-finding by descent.

At  $k = 29$  with symmetry breaking: 25,799 variables, 91,426 clauses.

## III. Instances and empirical hardness

An instance is determined by (output tensor of a scheme representative,  $k$ ). Tensors come from the public database and the record chain; the de Groote action makes the output form-set depend only on the pair  $(R, P)$  of  $\text{GL}(3, 2)$  matrices; each of the  $168 \times 168 = 28,224$  pairs is a cell (one concrete output-side instance), so every class supplies  $\sim 28k$  cells — a practically unlimited supply, with  $k$  the hardness dial:

- SAT phase ( $k \geq \text{minimum}$ ): minutes at minimum + 1; witnesses are extracted and replay-verified by the generator.
- Deep UNSAT ( $k$  well below minimum): easy.
- Boundary ( $k \in \{\min - 1, \min\}$ ): hard. On the calibrated seed cells no tested configuration decides the boundary within 10–30 minutes (Apple M4 Pro, one core per solver): kissat (600s and 1800s), CaDiCaL (600s), CryptoMiniSat 5 (615s, plain and native-XOR), Z3 4.16 on a word-level QF\_BV formulation (600s), and a kissat+CaDiCaL+CMS portfolio with symmetry breaking (900s).

TABLE I  
Calibrated seed cells (SAT witnesses verified).

| cell                              | weight | GF(2) min        |
|-----------------------------------|--------|------------------|
| sun56 output side (record scheme) | 49     | $\in \{29, 30\}$ |
| cn120 output side ( $C=28$ rep)   | 60     | $\in \{27, 28\}$ |

Deciding a boundary certifies a per-class optimum: e.g. an UNSAT certificate at the transposition-computed minimum proves that a class’s output side cannot be smaller, and (with the input-side exhaustions) that its total is optimal — a DRAT-checkable optimality theorem for a fixed class, now that a 55 scheme is known and the target is 54.

#### IV. Proposed benchmark set

Twenty instances spanning the phases: the two seed cells at  $k \in \{\min - 1, \min, \min + 1\}$ ; the identity cells of the two other known 56-addition classes (i12w219c23ci, i19w225c4efh) at their boundaries; and four fat-sides window cells of the record class at  $k = 27$ ; each with and without symmetry breaking. All are emitted by the C-side lower-bound generator (cxlb.py):

```
python3 matmul/cxlb.py --bits SCHEME.bits \
--k K --dump out.cnf
python3 matmul/cxlb.py --bits SCHEME.bits \
--k K --dump out.xnf # native XOR
```

(–no-sb disables symmetry breaking; all scheme bits files are committed in the repository.)

#### V. Availability

Generator, scheme inputs, verification tooling, and research notes: <https://github.com/gsidebottom/logic>. Submitted instances may be used freely under the competition’s standard terms.

#### References

- [1] Y. Sun. An Exact 56-Addition, Rank-23 Scheme for General  $3 \times 3$  Matrix Multiplication. arXiv:2604.27645, 2026.
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